CAPSTONE PROJECT PROPOSAL

MoodBand: An Inclusive Wearable System for Real-Time Mood Detection and Adaptive Sensory Feedback

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Abstract

Emotional well-being is closely tied to mental health, productivity, and quality of life, yet many people — particularly those who are visually impaired — lack accessible, unobtrusive tools to monitor and regulate their own mood in real time. This proposal presents MoodBand, a system that tests whether a user's mood can be reliably predicted from physiological signals alone. It reads heart rate and step data from a Fitbit wearable via the Google Health API, classifies the wearer's mood (happy, neutral, sad, stressed, or angry) with a lightweight machine-learning model that uses only physiological and motion signals, and responds through a companion mobile app on Android and iOS. When a negative mood is detected with enough confidence, the app offers a short intervention — for example, a slow vibration-paced breathing exercise for stress or a gradually brightening music sequence for sadness — delivered through visual, audio, and haptic channels chosen according to the user's accessibility profile (sighted, low vision, or blind). Intervention timing follows a just-in-time adaptive design with guardrails against acting on uncertain predictions, and the user is always asked before an intervention begins. On the public LifeSnaps Fitbit dataset, a Random Forest reaches about 0.33 balanced accuracy across four mood classes on people it has not seen, against a 0.25 chance level; neither more complex models nor combinations of models improved on it in a usable way. The feedback-decision module and the companion app have been built and tested together on recorded data, and the Fitbit has arrived for live integration. If physiological signals prove reliable enough, the next step after this project is a custom wearable. This document outlines the background, problem statement, objectives, literature review, system design, methodology, timeline, and budget for the project.

1. Introduction and Background

Wearable technology has moved from simple activity tracking toward richer physiological sensing, enabling continuous, passive insight into a user's physical and emotional state. Emotions influence decision-making, social interaction, and long-term mental health, yet most people have no simple way to notice a negative mood shift before it escalates into stress or withdrawal. Existing mood-tracking apps largely rely on manual self-reporting, which is easy to forget and inherently reactive rather than real-time.

Visually impaired individuals face an additional barrier: most consumer mood-tracking and well-being tools are built around visual interfaces — screens, color-coded charts, and icon-based dashboards — that are inaccessible without a screen reader, and screen readers do not reliably make up for this in practice. A system that senses mood from an existing wearable such as a Fitbit and responds through sensory channels chosen for each user can serve sighted and visually impaired users from the same platform, rather than requiring a separate accessible version as an afterthought.

2. Problem Statement

Current mood-monitoring solutions share three limitations that this project seeks to address:

- They are largely self-report based, requiring conscious, repeated user input rather than continuous passive sensing.
- They provide little or no closed-loop intervention — most only log or visualize mood after the fact, without acting to help the user regulate it in the moment.
- They are not designed with accessibility in mind, offering feedback almost exclusively through visual dashboards that exclude visually impaired users.

There is therefore a need for a wearable-integrated system that detects negative mood states in near real time and responds, at an appropriate moment, with feedback suited to the user's sensory needs, with the goal of helping the user toward an improved emotional state.

3. Aim and Objectives

3.1 Aim

To design, develop, and evaluate a mood-detection system that uses a Fitbit wearable to sense a user's physiological signals, classifies their emotional state in near real time, and delivers adaptive auditory, haptic, or visual feedback through a companion mobile app, for both sighted and visually impaired users, on both Android and iOS. A central question is whether mood can be reliably predicted from physiological markers alone. If it can, the next step after the capstone is a custom wearable.

3.2 Objectives

1. Review existing literature on affective computing, wearable physiological sensing, mood classification, accessible feedback, and mood-regulation interventions.
2. Integrate with a Fitbit wearable via the Google Health API to retrieve heart-rate and motion data in near real time, on both Android and iOS.
3. Train and benchmark machine-learning models that classify mood into five categories (happy, neutral, sad, stressed, and angry) from physiological and motion signals only, evaluated on people the model has not seen.
4. Design a feedback-decision module that decides whether and how to intervene, based on the predicted mood, its confidence, recent context, and the user's accessibility profile.
5. Implement auditory, haptic, and visual feedback in the companion app, accessible through screen readers.
6. Evaluate classification performance on public datasets and the project's own data, and evaluate the feedback with volunteer participants in a small user trial.

4. Scope of the Project

MoodBand implements a four-stage sense → process → classify → feedback loop: physiological and motion data are collected from a Fitbit wearable via the Google Health API; that data is processed and classified into a discrete mood by a trained machine-learning model; the classified mood is used to decide whether to intervene and with what; and the chosen feedback is delivered to the user through their phone, closing the loop. Fitbit was selected over alternatives such as the Apple Watch because its cloud-based API works identically regardless of the phone's operating system, so the companion app can support both Android and iOS users.

The sense-to-feedback loop is a logical pipeline, not a mandated physical deployment. The processing (mood classification) and feedback-decision stages may run on the mobile device, on a connected laptop, or on a cloud backend, whichever proves most practical given on-device performance constraints. A fully on-device implementation is the ideal target, but not a strict requirement; what matters is that the complete loop works end-to-end in near real time. The feedback-actuation stage (speaker, vibration motor, screen, and notifications) necessarily runs on the user's phone.

The scope includes detection of a limited set of discrete mood categories rather than a continuous emotional spectrum, and evaluation with a small user trial of volunteer participants, appropriate for a capstone project completed by a single student. The trial will proceed only once the supervisor has confirmed the user-trial (consent) agreement.

Design or fabrication of custom wearable hardware is outside the scope of this project, since an existing consumer device is used for sensing instead; a custom wearable is the intended next step if mood can be reliably predicted from physiological signals. Also outside of scope: long-term clinical validation, mass-manufacturing considerations, and integration with health platforms beyond the Google Health API.

A related but distinct idea — teaching visually impaired users to interpret colour-based mood cues directly through a dedicated non-visual training modality — is deliberately kept out of scope. Within this project, visually impaired users are served by the auditory and haptic feedback channels described in Section 8. The colour-decoding concept is proposed as a separate, follow-on project.

5. Literature Review

5.1 Affective Computing Fundamentals

Picard's foundational work [1] is where affective computing as a discipline really starts. Her central claim was that computers could be built to recognize and respond to emotional states, not just ignore them as noise. That idea sounds almost obvious now, but at the time it wasn't, and it's still the premise MoodBand rests on: that internal affective states leave measurable traces a system can learn to read.

Decades later, Conceição et al. [2] show what that premise looks like in practice with modern tools. They build on Russell's circumplex model, which maps emotion along two axes, valence and arousal, and use a deep-seeded clustering algorithm to classify emotional states from signals collected by a wearable wristband, testing it across three separate benchmark datasets. What makes this paper useful for MoodBand specifically is that it's working with the same kind of hardware form factor (a wrist-worn device) and the same underlying signal types we'd expect from a Fitbit, and it still manages to map those signals onto discrete emotional categories reasonably well.

Implication for MoodBand: together, [1] and [2] justify the project's core premise — that physiological signals can be mapped onto discrete emotional categories — and specifically justify grounding MoodBand's mood labels (happy, sad, stressed, angry, neutral) in the valence/arousal circumplex model rather than inventing an ad hoc labelling scheme. This directly shaped the mood categories used throughout the system design.

5.2 Wearable Physiological Sensing

Moving from theory to the specific question of wearables, Saganowski et al.'s systematic review [3] is a useful starting point. They find that heart rate, HRV, and electrodermal activity dominate the literature as the go-to signals for everyday emotion recognition, and they flag how much harder things get once you move out of the lab and into real, everyday conditions. That's exactly the setting MoodBand is targeting.

On the consumer-hardware side specifically, [4] is probably the single most reassuring paper on this list. Using the LifeSnaps dataset, built from actual Fitbit Sense smartwatches rather than research-grade sensors, the authors predict anxiety from everyday wearable data and get results suggesting a plain consumer device carries useful signal, not just noise.

That said, [5] is a good reality check on the same point. The authors ran into some genuinely annoying constraints when pulling Fitbit data for ML purposes: step counts reading zero while the device is just charging, HR data missing when the device isn't worn properly, all requiring manual filtering. Most notably, Fitbit only computes HRV while the user is asleep and relatively still, which means HRV cannot be relied upon as a live, daytime signal.

Implication for MoodBand: papers [4] and [5] both use Fitbit devices specifically, which is directly relevant since Fitbit is also the device selected for this project. The choice of Fitbit was driven primarily by a requirement outside the scope of this literature review on its own: the companion app must work for both Android and iOS users. Fitbit's cloud-based API serves data identically regardless of the user's phone platform, whereas device-native alternatives such as Apple's HealthKit only work on iOS. This means the HRV constraint identified in [5] — that Fitbit only computes HRV overnight — carries over directly into MoodBand's design (to be confirmed under the new API; Section 7.1), and the classification pipeline is built around that constraint rather than trying to design it away.

5.3 Mood/Emotion Classification Models

Once the signals are available, the question is what model actually does the classifying. The systematic review in [6] gives a pretty clear answer: classical machine learning methods, Random Forest and Support Vector Machines in particular, show up in the large majority of wearable-based stress detection studies, with Random Forest especially holding up well on EDA and HRV features.

[7] adds two things on top of that. First, the authors design their classification approach specifically for resource-constrained, on-device deployment — directly relevant given MoodBand aims to run its classifier on the phone where feasible. Second, they compare personalized models against generalized ones and find personalized

models can win out even with limited data per person, which matters given MoodBand's own user trial will necessarily be small.

Implication for MoodBand: [6] directly influenced the choice of classifier family — Random Forest is used as the primary candidate classifier specifically because it is the best-performing classical model on EDA/HRV-style wearable features in the reviewed literature, rather than defaulting to a heavier deep-learning approach that would be harder to justify running on a phone. [7] influenced two further decisions: favouring lightweight, resource-constrained model architectures suitable for on-device or near-device inference, and adopting a personalized-model evaluation strategy alongside a generalized one, given the project's necessarily small participant sample.

5.4 Accessible Feedback Design

The last piece of MoodBand, getting feedback to the user through non-visual channels, is backed by a smaller but still useful set of papers. The iFeeling system [8] is about as close a precedent as exists: it takes a detected emotional state and renders it as a distinct vibration pattern on a phone, built explicitly with visually impaired users in mind.

[9] widens the lens a bit. It's a recent, large-scale systematic literature review spanning two decades (2004 to 2024) of haptic assistive tools for blind and low-vision users, drawing on 132 papers, and it's useful less for any single finding and more as a way to check iFeeling's design choices against the broader literature.

Finally, [10] grounds all of this in actual user experience. Yeo and Chew interviewed visually impaired people in Singapore about how they use mobile apps day to day, and the picture isn't entirely rosy: screen readers often don't work as reliably as one would hope. Their recommendations explicitly call for combining audio and haptic feedback rather than relying on either alone, and for taking users' emotional wellbeing seriously as a design goal.

Implication for MoodBand: [8] directly shaped the haptic feedback channel — MoodBand's haptic channel builds on iFeeling's precedent of conveying affect through distinct phone-vibration patterns. [9] broadened this into a fuller design vocabulary: MoodBand's use of graduated or directional vibration patterns (e.g., descending vibration intensity for anger) draws on patterns documented across this wider body of research rather than the single iFeeling precedent alone. [10] most directly shaped a specific design rule: because screen readers were found to behave unreliably in practice, MoodBand pairs every visual cue with a corresponding audio or haptic equivalent by default, rather than treating a screen-reader-compatible UI as sufficient on its own.

5.5 Mood-Regulation Interventions and Feedback Delivery

The sections above cover how to detect a mood and how to deliver feedback non-visually, but not which intervention actually helps with a given mood, or when it's appropriate to say anything at all. That gap matters more than it might first appear: MoodBand's current mood classifier, a Random Forest trained on heart rate and step data from the 30 minutes before each self-reported mood in the LifeSnaps dataset, reaches about 0.33 balanced accuracy against a 0.25 chance baseline across four classes. That's real signal, but modest enough that acting on every prediction as if it were certain would be a mistake. This section looks at what the literature says about choosing and timing interventions, with that constraint specifically in mind.

Nahum-Shani et al. [11] provide the organizing framework. They define a just-in-time adaptive intervention (JITAI) by six components: a distal outcome, proximal outcomes, decision points, intervention options, tailoring variables, and decision rules — with tailoring variables capturing either vulnerability/opportunity (is intervention needed right now) or receptivity (can the person actually engage with it right now). It's a structure built for exactly this kind of problem: a system that has to decide, repeatedly, whether and how to intervene based on an imperfect read of someone's state.

Implication for MoodBand: [11] directly shapes how the feedback-decision module is framed — not as a simple mood-to-feedback lookup table, but as a JITAI decision rule, where each new prediction is a decision point, mood and context (e.g., recent activity) are the tailoring variables, and the feedback catalog is the set of intervention options.

Several papers speak directly to what goes wrong when that decision rule is too eager. Koh et al.'s systematic review [18] screened 2,152 studies down to 52 on stress-tracking and intervention systems, and found just-in-time delivery

to be the most common approach (30 of 52), with on-demand delivery close behind (24). More usefully for MoodBand, it also catalogs what tends to fail: interventions delivered at unwanted moments cause fatigue and attrition, a "perceptual mismatch" between what the system flags and what the user is actually experiencing (their example: a user who feels excited but is flagged as stressed because of high arousal) erodes trust, and detection on its own explains what state someone is in but not why. Its main recommendation is human-in-the-loop calibration — letting users correct the system's read of their state. A pilot study on wearable-based self-management during an endurance-cycling program [19] (a preprint, not yet peer-reviewed) backs this up anecdotally: participants generally trusted the detection but some found the alerts "overly sensitive," and preferred brief, easy-to-interpret notifications. A University of Twente bachelor's thesis [21] — a weaker, non-peer-reviewed source, flagged here as such — adds a related caution about how feedback is worded: among 26 participants who alternated a week with and a week without a stress-feedback smartwatch, the watch had no significant effect on perceived stress or health anxiety overall, but in interviews some participants described feeling more stressed by it.

Implication for MoodBand: given MoodBand's own classifier accuracy is modest, [18] and [19] together justify building several guardrails into the feedback-decision module rather than treating every prediction as actionable. Concretely: the system only acts when the model is confident and the mood has persisted across several recent predictions, rather than on a single reading; interventions are rate-limited (a default 30-minute cooldown, a daily cap — one system in [18]'s review capped interventions at four per day — quiet hours, and a user-controlled snooze); a detected "stressed" state is suppressed if recent step data shows the user has just been physically active, both to avoid the perceptual-mismatch failure [18] describes and in line with systems in [18]'s review that used accelerometer data to filter out periods of intense movement that distort physiological signals; feedback is framed as a suggestion ("Want a 1-minute breathing break?") rather than a diagnostic claim ("You are stressed"), since [21] suggests that for some people the feedback itself can add stress; and every intervention carries a simple "Was this right?" (yes/no/snooze) response, which both respects [18]'s human-in-the-loop recommendation and doubles as in-the-moment labelled data for the personalized-model strategy already motivated by [7]. An on-demand "I need a break" option is offered alongside these automatic triggers, reflecting [18]'s finding that on-demand delivery is nearly as common as just-in-time delivery in practice.

For the stress intervention itself, several lines of evidence point toward a breathing-based response. Two meta-analyses reach somewhat different conclusions on how strong the effect is: Fincham et al. [12], pooling 12 randomized controlled trials and 785 adults, found breathwork reduced self-reported stress versus controls ($g \approx -0.35$, with secondary reductions in anxiety and depressive symptoms); Shao et al. [13], a larger pool of 31 studies and 1,133 non-clinical participants, found clear cardiovascular effects from slow-paced breathing but only a marginal effect on perceived stress (SMD ≈ -0.51, $p = .06$) — a useful reminder that the physiological and the self-reported effect aren't always the same size. On the delivery side, a small set of HCI studies suggest a wearable can actually guide this without a screen: Miri et al.'s personalized vibrotactile breathing pacer [14] reduced anxiety under a cognitive stressor relative to a control ($d \approx .33$, $p = .004$), building on the same group's earlier design study [22], which found that personalization mattered and concluded that strong-exhale-phased vibration patterns were the most suitable choice; and Costa et al.'s EmotionCheck [15] found, in a lab experiment with 67 participants, that slow wrist vibrations at 60 beats per minute, which participants were told reflected their own heart rate, kept anxiety lower than in a control group, even though the feedback was false. A field study with law-enforcement officers [20] adds a practical data point on a smartwatch that prompted breathing or meditation when heart rate stayed elevated: 22 officers wore the watch for 30 days and 18 were interviewed afterwards; the vibration notification itself was among the three components they found most useful, though receipt of notifications varied widely and only half recalled ever feeling one. Separately, de Witte et al.'s meta-analysis of 104 RCTs and 9,617 participants [16] found music interventions reduce stress-related outcomes generally (physiological $d \approx .38$, psychological $d \approx .55$), with the single largest physiological effect specifically on heart rate ($d \approx .46$).

Implication for MoodBand: together, [12]–[15], [20], and [22] are why the stressed-mood intervention is a slow breathing pacer at about six breaths per minute — the rate Shao et al. [13] describe as typical of slow-paced breathing, to be tuned in the user trial — delivered by default as a strong-exhale-phased vibration pattern, the design [22] recommended for the PIV pacer [14], with a slow, heartbeat-like tap offered as an alternative — both patterns directly modelled on what [14] and [15] found to work without a screen, though both used body-worn devices ([22] recommended placing the pacer on the abdomen, and [15] was worn on the wrist), and whether the same patterns work from a phone's vibration motor is something MoodBand's trial will need to test. [16] is why music is offered as the audio option for stress specifically, rather than an arbitrary playlist: among the physiological outcomes [16] measured, heart rate showed the largest effect — the very signal MoodBand's sensing relies on. And because

MoodBand's “stressed” class comes from LifeSnaps' “Tense/Anxious” label, [23]'s finding that playlist ordering made no difference after an anxiety induction suggests the stress music option doesn't need an iso structure.

Sadness is covered by a single, smaller study. Starcke et al. [17] induced sadness experimentally and then played participants one of several music sequences; a sequence that started sad and gradually became happier (the “iso principle”) left people with higher positive affect and valence and lower negative affect than the comparison sequences, though not every pairwise difference reached significance and the sample was small. A larger preregistered experiment adds a note of caution: Lowe-Brown et al. [23] found no significant difference between iso-principle and compensatory playlists among 193 healthy participants after an anxiety induction. That study targeted anxiety rather than sadness, so it doesn't contradict [17], but it is a reason not to assume the iso principle works across emotions.

Implication for MoodBand: [17] is the reason the sad-mood audio intervention is an iso-principle sequence — music that matches the user's current low mood first, then gradually brightens — rather than the uplifting playlist originally proposed. Given this rests on one small study, and [23] found no iso advantage for anxiety, it's treated as a reasonable starting point to test in the user trial, not a settled finding.

A related question is when to say nothing at all. Koh et al. [18] found that interventions delivered at unwanted moments cause fatigue and attrition. By the same logic, repeatedly confirming a happy or neutral state risks the same fatigue without any regulatory benefit.

Implication for MoodBand: that inference from [18] is why happy and neutral states trigger no feedback at all while they continue, a departure from the original proposal's plan of a green confirmation banner or pleasant chime for these states.

Two other planned behaviors have no direct support in the sources reviewed here, and that's worth being explicit about rather than implying otherwise. No source addresses anger-specific interventions, so the calming-timer response kept for anger remains a design choice carried over from the original proposal — to be evaluated in the user trial rather than assumed correct, and relevant only once the classifier itself supports an anger label. Similarly, no source here speaks to whether positive reinforcement should be delivered specifically on recovery from a negative state to a neutral or positive one; restricting feedback to transitions rather than steady states is motivated by the fatigue concern above, not by direct evidence that reinforcement itself helps, and is likewise something to check empirically rather than assume.

Finally, the question of which channel to use for which user is already addressed by the accessible-feedback literature in Section 5.4, but two of the stress papers reinforce it from a different angle: both Miri et al.'s pacer [14] and Costa et al.'s EmotionCheck [15] regulate affect through vibration alone, with no visual or auditory component at all, which is direct evidence that haptic-only regulation is viable rather than a fallback for when a screen isn't usable — though neither study included visually impaired participants, which MoodBand's user trial will need to check.

Implication for MoodBand: combined with [8], [9], and [10] from Section 5.4, this is why MoodBand defines three accessibility profiles rather than a single sighted/non-sighted split: sighted users get visual feedback with haptic by default; low-vision users get audio and haptic, with high-contrast visuals available optionally; and blind or visually impaired users always get audio and haptic together, never one alone. Every visual cue keeps a non-visual equivalent across all three profiles.

5.6 Synthesis

Put together, the pieces MoodBand depends on each have reasonable support behind them. Physiological signals can be mapped onto discrete moods [1], [2]. A consumer wearable like Fitbit is a workable, if imperfect, way to collect those signals [3]–[5]. Classical, lightweight models are a sensible fit for the classification step given the deployment constraints [6], [7]. Haptic feedback in particular has real precedent for communicating affective state to visually impaired users [8]–[10]. And a growing body of intervention research supports specific, non-visual responses to at least two of MoodBand's five target moods: breathing-based feedback and music for stress [12]–[16], [20], [22], and iso-principle music for sadness [17], though a larger study found no iso advantage for anxiety [23]. These sit within a just-in-time adaptive framework [11], with guidance on when not to intervene [18], [19], [21]. Two gaps remain: no evidence here addresses anger-specific interventions or positive reinforcement on recovery, so those are carried

as explicit design choices. What doesn't exist yet, as far as this review found, is a single system that ties all five pieces together into one mobile-first pipeline — a gap this project aims to address.

5.7 Sensor Recommendations Across Reviewed Literature

Table 1 maps each reviewed paper to the physiological sensors or signals it used or recommended, alongside the sensors MoodBand will actually move forward with.

Paper	Sensors Used / Recommended
[1] Picard (1997)	N/A — conceptual/foundational text, no specific sensor
[2] Conceição et al.	Heart rate / blood-volume pulse (PPG), electrodermal activity (EDA), skin temperature (via Empatica E4 wristband)
[3] Saganowski et al. (review)	Heart rate, HRV, electrodermal activity (identified as the dominant signals across the wearable-sensing literature)
[4] Velu et al. (LifeSnaps/Fitbit)	Heart rate, Fitbit-derived stress score, sleep, step count (Fitbit Sense)
[5] King et al.	Heart rate, HRV (sleep-only), step count / motion (Fitbit intraday data)
[6] Pataca et al. (review)	Electrodermal activity, HRV (found Random Forest performs best specifically on these two)
[7] Irfan et al.	ECG (minimal wearable sensing)
[8] iFeeling	N/A — haptic output/rendering paper, not a sensing paper
[9] Jiang et al. (review)	N/A — haptic feedback review, not sensing
[10] Yeo & Chew	N/A — qualitative UX study, not sensing
Sensors MoodBand will use	**Heart rate (continuous, real-time) and motion/accelerometer (continuous), with overnight heart-rate variability (HRV) tested as a supplementary signal; it lowered accuracy for people the model hadn't seen, so the general model doesn't use it**

Table 1: Sensors recommended per paper vs. sensors selected for MoodBand

These are the signals consistently available through a consumer-grade, cross-platform-friendly wearable API (Fitbit, via the Google Health API). Electrodermal activity, used in the lab-grade Empatica E4 wristband in [2] and recommended generally in [3] and [6], is out of reach because the Inspire 3 has no EDA sensor. Skin temperature is available from the Google Health API only as a nightly reading, not as a continuous daytime signal, so it cannot serve as a real-time feature; as a nightly feature it did not improve accuracy (Section 7.2). ECG-based sensing as explored in [7] is also out of reach: the Google Health API has an ECG data type for Fitbits with an ECG app, but the Inspire 3 has no ECG sensor, and ECG on those devices is a manual spot check rather than a continuous signal. MoodBand's sensor selection therefore reflects the intersection of what the literature recommends and what a consumer-grade, cross-platform wearable actually provides, with HRV, which [5] found is computed only overnight (to be confirmed under the new API; Section 7.1), tested only as a supplementary signal.

5.8 Device Selection Summary

To select a specific wearable, four candidate devices — Fitbit, Apple Watch, Samsung Galaxy Watch, and WHOOP — were compared against price, cross-platform support, developer API accessibility, real-time data availability, and development-environment fit. This section summarizes that comparison and its conclusion.

One piece of context shaped the timing of this decision: the legacy Fitbit Web API (the one referenced by [4] and [5]) was retired by Google and replaced by the new Google Health API: developer support ended on September 30, 2026, and the API was turned off in early November 2026. This is a net positive for the project: the legacy API

required a justification request to unlock sub-daily heart-rate data, whereas the new API provides intraday heart rate by default with no special approval step, stored at up to 1-second resolution; the Inspire 3 records heart rate every 5 seconds, and every second during exercise.

Criterion	Fitbit	Apple Watch (SE 3)	Samsung Galaxy Watch	WHOOP
Price (new / used)	$70–130 / $30–80	$220–250 / $120–180	$170–310 / $100–150	Bundled with membership
Works on Android?	Yes	No — iPhone-only, structurally	Yes	Yes
Works on iOS?	Yes	Yes	No — continuous streaming is Android-only	Yes
API approval needed?	No (new Google Health API)	No (HealthKit)	Only for Samsung's cloud API (1–2 wks)	Free API, but returns no data without paid membership
Real-time daytime HR?	Yes — 5-sec resolution	Yes (workout mode)	Yes — on-device Sensor SDK	No — no continuous/raw HR via API
Matches lit. review?	Yes — [4], [5] used Fitbit directly	No direct precedent	No direct precedent	No direct precedent

Table 2: Condensed device comparison

Decision: Fitbit, integrated via the Google Health API, was finalized as the project's sensing device. The deciding factor was the hard requirement that the companion app support both Android and iOS: this structurally rules out Apple Watch (pairs with iPhone only) and Samsung Galaxy Watch (continuous real-time streaming is Android-only via the Samsung Health Sensor SDK), regardless of their other technical merits. Fitbit's cloud-based API is the only option on the shortlist that serves identical data regardless of phone platform. WHOOP was ruled out independently of the platform question, since its API exposes no continuous or raw heart-rate data under any circumstance and requires an active paid membership just to return data at all.

6. Proposed System Overview

MoodBand is organized into four functional layers that implement the sense → process → classify → feedback loop described in Section 4. The sensing layer pulls heart-rate and motion data from the user's Fitbit via the Google Health API. The processing layer cleans the signal and classifies mood using a trained model. The decision layer decides whether to intervene and, if so, which intervention to offer, based on the predicted mood, its confidence, recent context, and the user's accessibility profile. The feedback layer delivers the intervention on the phone: first as a notification carrying a suggestion, then, once the user opens it and agrees, as a short session combining visual, audio, and haptic cues.

The processing and decision layers can run on the mobile device, a connected laptop, or a cloud backend; full on-device execution is the target end state, not a requirement. The feedback layer necessarily runs on the phone, since it drives the hardware the user directly experiences. Because the sensing layer uses Fitbit's cloud API rather than a platform-specific framework, the architecture works identically on Android and iOS. Figure 1 illustrates this architecture.

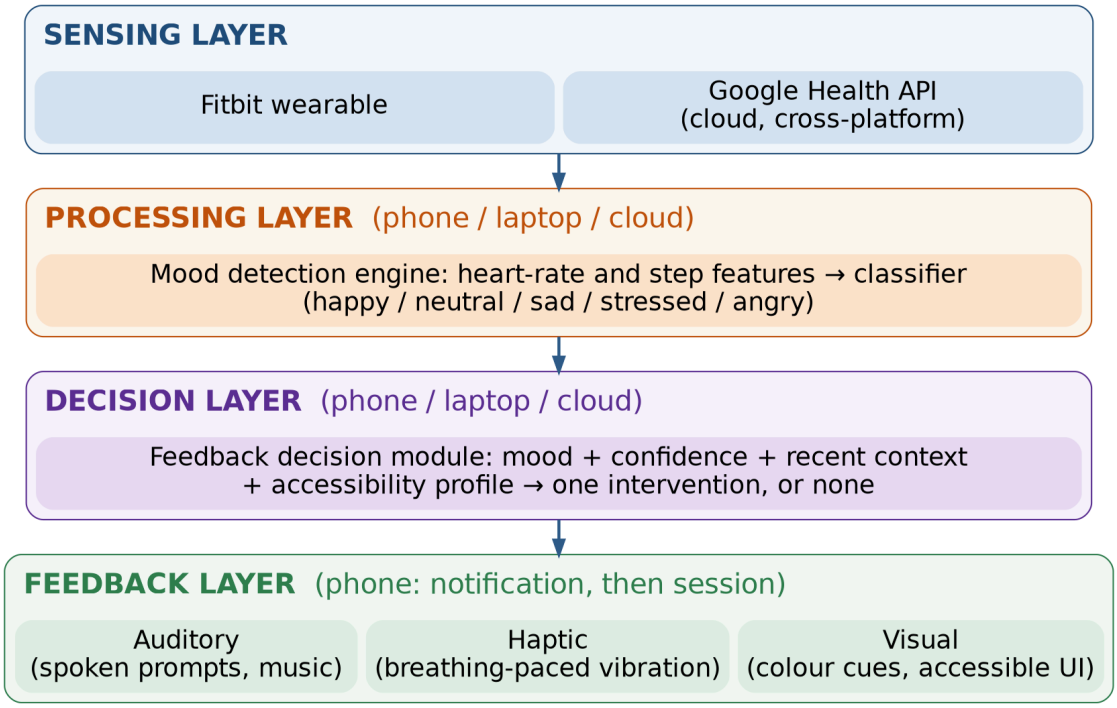


Figure 1: Proposed system architecture of MoodBand

7. Methodology

Because this is a single-student project with a fixed 16-week timeline, the system is built and validated in two phases. In the first phase, the mood-classification model, the feedback-decision module, and the mobile app are built and tested end-to-end using public wearable datasets as a stand-in for live sensor data, so that design problems surface early. Only once this offline version works does the second phase begin: integrating the live Fitbit sensing layer via the Google Health API, followed by the user trial. This ordering is reflected in the timeline (Section 11).

7.1 Fitbit Integration

MoodBand integrates with a Fitbit Inspire 3 via the Google Health API, the successor to the legacy Fitbit Web API, which Google retired in 2026. Fitbit was chosen because its cloud-based API returns the same data whether the companion app runs on Android or iOS. The Inspire 3 has an optical heart-rate sensor and a 3-axis accelerometer, which provide the two signals the model uses; the device, purchased by the supervisor, arrived in November.

A prototype against the Google Health API, run from a laptop, has started. Sign-in works end to end, and data calls succeed once the Google account has a Fitbit profile. The documentation adds three findings. The API stores heart rate at up to 1-second resolution, while the Inspire 3 records it every 5 seconds outside exercise, according to Google's specifications. The documentation does not say whether HRV is recorded only during sleep. And data reaches the API only after the band syncs through the Fitbit app, so "real time" means as recent as the last sync. While the project's Google sign-in is in testing mode, up to 100 listed users can sign in, and each must sign in again weekly; this is enough for the user trial.

Still to be checked with the band's own data: how often it actually samples heart rate, whether HRV appears only overnight, how far the data lags behind real time, sign-in from the phone app, and whether a MoodBand notification mirrored to the band can be felt on the wrist. Any gap will be documented and the design adjusted.

7.2 Data and Mood Classification

Model development uses public datasets. The primary dataset is LifeSnaps (Fitbit Sense, 71 participants), whose in-the-moment mood reports are mapped to MoodBand's classes: Happy, Neutral, and Sad map directly, and Tense/Anxious maps to stressed. For each report, heart-rate and step features are computed from the 30 minutes before it, using minute-level data, which mirrors how the app will classify from a user's most recent Fitbit data. This gives 1,961 usable reports from 61 participants (neutral 36%, happy 32%, stressed 26%, sad 7%).

The current Random Forest reaches a balanced accuracy of about 0.33 on participants not seen in training (participant-grouped cross-validation, averaged over five seeds), against 0.25 chance. Sad remains the hardest class (F1 about 0.13). Personalized models scored higher (about 0.38–0.39). Personalization is kept as a layer on top of the general model, and its results are reported separately, so that the headline result reflects the physiological signals alone. Adding last night's HRV and sleep lowered accuracy on unseen participants, so these features are kept only as an option for personalized models. A cross-dataset check on PMData was inconclusive because its labels are once-daily wellness scores rather than in-the-moment moods, so PMData is not used for training.

Because the project's central question is whether physiological signals alone can predict mood, the model uses physiological and motion signals only: heart rate, steps, and per-person physiological baselines such as the person's own average heart rate. It never uses a person's past mood reports or answers as inputs. Personalization is a separate layer on top of this general model, made up of personal heart-rate and activity baselines, retraining on the user's own labelled moments, and personalized feedback. The headline result is therefore the physiological-only model's balanced accuracy on people it has never seen, which is what a future custom wearable could expect from these signals.

As the supervisor requested, the Random Forest was benchmarked against five alternatives on identical participant-grouped splits (Table 3). An RBF SVM scored slightly higher (0.351 against 0.334), mainly by detecting sad more often, but its macro-F1 was the same, and the more complex models (gradient boosting and a small neural network) did worse. With about 2,000 labels and 12 features, model capacity is not the bottleneck; the limited mood signal in heart rate and steps is. The SVM's lead also disappeared once it was made to produce the class probabilities the feedback module needs: with calibrated probabilities, its balanced accuracy fell to 0.320 and it almost never predicted sad or stressed, so in the feedback replay it suggested no breaks at all. The Random Forest therefore stays the model. A confusion matrix is kept for every model compared.

Model	Balanced accuracy	Macro-F1	Recall: happy / neutral / sad / stressed
SVM (RBF)	0.351 ± 0.002	0.326	0.38 / 0.29 / 0.32 / 0.42
Random Forest	0.334 ± 0.006	0.327	0.48 / 0.33 / 0.17 / 0.35
Logistic regression	0.316	0.271	0.31 / 0.16 / 0.38 / 0.42
Neural network (MLP)	0.311 ± 0.037	0.269	0.26 / 0.32 / 0.36 / 0.30
Gradient boosting	0.297 ± 0.005	0.295	0.46 / 0.38 / 0.04 / 0.30
Majority class	0.250	0.132	—

Table 3: Model benchmark on LifeSnaps (balanced accuracy averaged over five seeds; chance = 0.25)

Two further experiments looked for higher accuracy. Adding other signals the Google Health API exposes — activity-adjusted heart rate, longer context, calories, exercise sessions, and last night's HRV, SpO2, skin temperature, and sleep — or sign-up profile data did not beat the 12 baseline heart-rate and step features for unseen people. The overnight and profile features lowered accuracy by 0.02–0.05, because the models used them to tell people apart rather than to read their mood. Combining models, which the supervisor asked about at the mid-term review, did not help either: soft voting matched the Random Forest, and stacking gained only 0.007 while producing probabilities so flat that just 3% of predictions reached the 0.45 confidence gate, so the feedback module would almost never act. Reweighting the model by a person's own past mood mix would raise balanced accuracy to about 0.40–0.43, but controls show that most of that gain comes from knowing a person's usual mood rather than reading their body, so it is not used in the general model. On top of a person's mood mix, the heart-rate and step readings add 0.04–0.07,

which shows what the signals themselves carry. Retraining the model with a person's own reports weighted more heavily ("personal weighting") reached about 0.37–0.38; this is the method the personalization layer will use.

LifeSnaps has no anger label, but the final classifier must detect all five moods, so additional data is required. An access request has been submitted for the EmoPhone dataset (Fitbit Inspire HR), whose 2022 wave (D-3) includes anger ratings, and the student has followed up with its authors. TILES-2018 (Fitbit Charge 2) would add further stress and affect data; whether any of its items capture anger has not yet been confirmed, and it requires a data access agreement signed on behalf of Northeastern, which is with the supervisor. The benchmark will be re-run on these datasets once either is available.

7.3 Feedback Decision Logic

The feedback-decision module turns each mood prediction into either no action or one intervention. Following the just-in-time adaptive intervention framework reviewed in Section 5.5, each new prediction is a decision point, and the module applies a series of guardrails before acting:

- Confidence gate: act only if the top class's probability is at least 0.45 and at least 0.10 ahead of the runner-up.
- Persistence: the same mood must appear in at least two of the last three predictions.
- Exercise filter: a predicted "stressed" (or "angry") state is ignored when recent step data shows physical activity.
- Rate limits: a 30-minute cooldown between interventions, a cap of four per day, quiet hours, and a user snooze.
- No feedback for continuing happy or neutral states; a brief positive cue only on recovery from a negative mood.

Every intervention is phrased as a suggestion ("Want a 1-minute breathing break?") rather than a diagnosis. Opening it shows a first screen with three choices — Start, Not now, or Snooze for an hour — and nothing plays until the user chooses Start. Afterwards, the user is asked "Was this right?" (yes / no / snooze). The answers are logged as evaluation labels and as training labels for the per-user personalization layer; they are never used as inputs to the mood model. An on-demand "I need a break" option is available alongside the automatic triggers and starts immediately.

The module has been built and tested in Python. To choose the confidence threshold, held-out LifeSnaps predictions were replayed every 10 minutes from 08:00 to 22:00 across 1,295 participant-days (about 101,000 predictions). At 0.45, MoodBand intervenes about 1.2 times per person per day. Precision for "stressed" (an intervention within 30 minutes of a matching self-report) stays around 0.31–0.37 at every threshold tested, against 0.26 chance, showing that the model's confidence does not track correctness: the threshold controls how often MoodBand speaks up, not how often it is right. This is why suggestion framing, the go-ahead step, and the "Was this right?" labels are essential parts of the design.

7.4 Companion Mobile Application

The companion app is built with React Native and Expo (as a development build), which renders native platform controls so that screen readers such as VoiceOver behave as in any native app. Haptic patterns, including the breathing pacer's gentle inhale and strong exhale, are played through the platform's haptics engine. The app lets users set their accessibility profile and feedback preferences, see their past answers, and start an on-demand break; a mood-history view will be added later.

A haptics prototype tested on an iPhone showed that the breathing pacer is clear and calming while the phone is held, but barely noticeable on a table and not noticeable in a pocket. The phone's operating system also stops an app's custom haptics while it is in the background. Automatic interventions are therefore delivered in two steps: a notification carrying the suggestion, with the system sound and vibration (which screen readers read aloud), followed by the session itself, which plays only after the user opens the notification and agrees to start. Whether Fitbit's notification mirroring can carry the alert to the wrist, where haptics are felt best, will be tested now that the device has arrived; the band cannot play custom patterns, so the breathing session itself still plays on the phone.

Before live Fitbit data is available, the app is connected to the real model and decision module through a replay mode. A laptop server plays a recorded LifeSnaps day through a Random Forest that never saw that participant and through the decision module, and the app plays the day at one hour per minute. Each suggestion is scheduled as a local notification, so it arrives even with the phone locked, and the tester's answers are saved alongside the participant's own mood report from that time. This completed the offline integration ahead of schedule and was tested on an iPhone. Because the tester is not the person whose data is replayed, their answers test the flow rather than the model; the participant's own reports remain the accuracy check. The live version will need push notifications for suggestions sent from a server, which requires the Apple Developer Program already included in the budget.

7.5 Testing and Evaluation

The headline evaluation is the physiological-only model's balanced accuracy on people it has never seen, compared against chance. Personalized results are reported alongside the general model's, not in place of them. Classification will be evaluated with participant-grouped cross-validation, using balanced accuracy and macro-F1 rather than plain accuracy because the mood classes are imbalanced, and with a confusion matrix for every model and dataset. Results will be averaged over several random seeds, since single runs vary by several points. The feedback-decision module is evaluated by replaying recorded data, as described in Section 7.3. The app is tested on device against a checklist covering each haptic pattern, the notification-to-session flow, each accessibility profile, and screen-reader use.

Finally, a small user trial with volunteer participants recruited from the student's friends will collect Fitbit data and in-the-moment mood labels using a protocol modelled on LifeSnaps' self-reports. This provides out-of-sample validation of the classifier and an assessment of the feedback, and will begin only once the supervisor has confirmed the user-trial agreement and participants have given informed consent.

Because participants are recruited from the student's friends, the trial is not expected to include visually impaired users. This is a limitation: the aim and objectives emphasize visually impaired users, and the literature review notes several design choices, such as whether haptic-only regulation works for blind and low-vision users, that need checking with them. For this project, the low-vision and blind profiles are evaluated through on-device testing with screen readers and the accessibility checklist rather than with visually impaired participants, and testing with visually impaired users is left for future work.

8. Feedback Interventions and Accessibility Profiles

Table 4 summarizes the intervention MoodBand offers for each detected mood, and Table 5 the channels each accessibility profile receives. Every visual cue has a non-visual equivalent and screen-reader text, and all feedback is delivered through the user's phone.

Mood detected	Intervention	Basis
Stressed	Breathing pacer at about six breaths per minute (gentle 4-second inhale, strong 6-second exhale), as a strong-exhale vibration pattern by default, with a slow heartbeat-like tap as an alternative; music offered as the audio option	Breathwork and slow-breathing meta-analyses; vibrotactile pacer studies; music meta-analysis (Section 5.5)
Sad	Iso-principle music sequence: music matching the low mood first, then gradually brighter	One small experimental study; treated as a starting point to test
Angry	Calming timer	No direct evidence found; a design choice, active once the classifier supports anger
Happy / Neutral	No feedback while the state continues	Inferred from evidence on intervention fatigue

Mood detected	Intervention	Basis
Recovery (negative → positive)	Brief positive cue, delivered as a quiet notification or shown when the app is next opened	No direct evidence found; a design choice to evaluate

Table 4: Mood-to-intervention mapping

Accessibility profile	Feedback channels
Sighted	Visual and haptic by default
Low vision	Audio and haptic, with optional high-contrast visuals
Blind / visually impaired	Always audio and haptic together, never one alone; if audio is unavailable (silent mode), haptic plus a deferred audio cue, never visual-only

Table 5: Feedback channels by accessibility profile

9. Tools and Technologies

- Wearable integration: Google Health API (Fitbit; cloud-based, works identically on Android and iOS).
- Machine learning: Python, pandas, and scikit-learn (the Random Forest was benchmarked against an RBF SVM, logistic regression, gradient boosting, a small neural network, and voting and stacking ensembles, and remains the model); TensorFlow Lite considered for on-device inference.
- Feedback-decision module: Python, with the intervention catalog stored as JSON so the same table can be used by a Python backend or the mobile app; a small laptop server (Python standard library) feeds recorded data to the app in replay mode.
- Mobile application: React Native with Expo (development build), with haptics via react-native-haptic-feedback (Core Haptics on iOS).
- Optional backend: a lightweight cloud function (e.g., Firebase Cloud Functions) if on-device inference proves too heavy; local on-device storage with Firebase sync for mood history.
- Version control and documentation: Git/GitHub.

10. Expected Outcomes

- Mood-classification model: a physiological-only model that classifies five moods (happy, neutral, sad, stressed, angry), evaluated on people it has not seen and reported with confusion matrices for every model and dataset, as evidence of how well physiological signals alone can predict mood.
- Feedback-decision module: a tested implementation of the just-in-time decision rules, with its threshold chosen by simulation on recorded data.
- Companion mobile app: an Android and iOS app that connects to a Fitbit via the Google Health API and delivers auditory, haptic, and visual feedback according to each user's accessibility profile, with full screen-reader support.
- Evaluation report: classification results on public datasets and trial data, together with findings from the user trial.
- Documented architecture: a documented, extensible design that can serve as a foundation for future work.

11. Project Timeline

The project is planned over a 16-week period, summarized in Table 6. As described in Section 7, the model, feedback module, and mobile app are built and tested first using offline, publicly available datasets; Fitbit / Google Health API integration is scheduled after mobile app development, once the full product has been validated to work end-to-end. Offline integration (the model, feedback module, and app running together on recorded data) was completed ahead of schedule in October, and the mid-term presentation was given on October 28.

Activity	Wk 1–2	Wk 3–6	Wk 7–10	Wk 11–13	Wk 14–16
Literature review & requirements gathering	X				
Dataset acquisition & preprocessing (LifeSnaps; EmoPhone wave D-3 for anger ratings; TILES-2018 for additional stress/affect data)	X	X	X		
Mood-classification model training & benchmarking (offline, public datasets)		X	X	X	
Feedback module (audio/haptic/visual) development		X	X		
Mobile companion app development			X	X	
Offline system integration & testing (app + model + feedback, no live wearable)				X	
Fitbit / Google Health API integration & setup				X	X
End-to-end testing with live wearable data					X
User trial with volunteer participants					X
Evaluation, documentation & final report					X

Table 6: Proposed project timeline

12. Budget Estimate

As the project now integrates with an existing Fitbit wearable rather than building custom hardware, the required budget is minimal compared to the original prototype-based plan. Table 7 summarizes the estimated costs.

Item	Estimated Cost
Fitbit wearable (Inspire 3 recommended)	Purchased by the supervisor for the project
Google Health API access (successor to the Fitbit Web API)	$0 (free for development use)
Mobile app development tools (React Native, Expo)	$0 (free)
Cloud storage/sync (Firebase)	$0 (free tier, sufficient for project scale)

Item	Estimated Cost
Apple Developer Program, to distribute the iOS app to trial participants through TestFlight (a free Apple ID limits installs to a few devices and the app stops opening after 7 days)	$99/year
Miscellaneous (testing devices for both Android and iOS, participant incentives)	$50
Contingency (10%)	$15
Total Estimated Cost (excluding the Fitbit device)	**~$164**

Table 7: Estimated project budget

13. Significance of the Project

This project contributes to two growing fields simultaneously: wearable-integrated affective computing and inclusive assistive technology. Rather than treating accessibility as an add-on, MoodBand is designed from the outset to serve sighted and visually impaired users through a single adaptable platform, offering a model for how future consumer well-being apps might be built for inclusivity by default. By building on existing wearable hardware rather than a custom device, the project also demonstrates a practical, low-barrier path to deploying accessible affective computing using devices people already own. The project also tests whether mood can be reliably predicted from physiological signals alone; if it can, the next step is a dedicated custom wearable. More broadly, the prototype could serve as a foundation for applications in stress management, mental-health support tools, and assistive technology for independent living.

14. Conclusion

This proposal has outlined the motivation, objectives, methodology, and implementation plan for MoodBand, a system that integrates with a Fitbit wearable to detect mood in near real time, on both Android and iOS, and respond with accessible, well-timed feedback aimed at improving the user's emotional state. By combining wearable-sourced physiological sensing, lightweight machine learning, literature-grounded intervention design, and accessibility-conscious feedback, built and validated offline before live wearable integration, the project addresses a genuine gap in current well-being technology while remaining feasible for a single student to complete within a capstone project timeline and budget.

15. References

[1] R. W. Picard, Affective Computing. Cambridge, MA, USA: MIT Press, 1997.

[2] M. A. Conceição, A. Dubois, S. Haustein, B. Miranda, and C. L. Azevedo, “Deep-seeded clustering for emotion recognition from wearable physiological sensors,” arXiv preprint arXiv:2308.09013, 2023 (revised 2025).

[3] S. Saganowski, B. Perz, A. G. Polak, and P. Kazienko, “Emotion recognition for everyday life using physiological signals from wearables: A systematic literature review,” IEEE Trans. Affective Comput., vol. 14, no. 3, pp. 1876–1897, 2023.

[4] A. Velu, J. Ramesh, A. Ahmed, S. Ganguly, R. Aburukba, A. Sagahyroon, and F. Aloul, “Integrating Fitbit wearables and self-reported surveys for machine learning-based state–trait anxiety prediction,” Appl. Sci., vol. 15, no. 19, p. 10519, 2025.

[5] Z. D. King, M. Khalid, H. Yu, K. Shibuya, K. Zanna, M. Majd, R. L. Brown, Y. Shen, G. Kypriotakis, T. Vaessen, C. P. Fagundes, and A. Sano, “Machine learning-based context aware EMAs: An offline feasibility study,” arXiv preprint arXiv:2506.15834, 2025.

[6] A. O. Pataca, E. Zdravevski, P. J. Coelho, N. M. Garcia, M. Deryuck, C. Albuquerque, and I. M. Pires, “Use of machine learning for predicting stress episodes based on wearable sensor data: A systematic review,” Comput. Biol. Med., vol. 198, pt. A, p. 111166, 2025.

[7] M. Irfan, A. Nawaz, A. K. Bulbul, R. Klén, A. Subasi, T. Westerlund, and W. Chen, “Emotion recognition with minimal wearable sensing: Multi-domain feature, hybrid feature selection, and personalized vs. generalized ensemble model analysis,” in Proc. 2025 IEEE Int. Conf. Bioinformatics and Biomedicine (BIBM), 2025, pp. 5447–5453.

[8] S. ur Réhman and L. Liu, “iFeeling: Vibrotactile rendering of human emotions on mobile phones,” in Mobile Multimedia Processing: Fundamentals, Methods, and Applications, X. Jiang, M. Y. Ma, and C. W. Chen, Eds., Lecture Notes in Computer Science, vol. 5960. Berlin, Germany: Springer, 2010, pp. 1–20.

[9] C. Jiang, E. Kuang, and M. Fan, “How can haptic feedback assist people with blind and low vision (BLV): A systematic literature review,” arXiv preprint arXiv:2412.19105, 2024.

[10] P. H. J. Yeo and K. H. Chew, “A framework for inclusive mobile application design: Insights from Singapore's visually impaired community,” Int. J. Asia Digit. Art Des., vol. 29, no. S1, pp. 46–56, 2025, doi: 10.20668/adada.29.S1_46.

[11] I. Nahum-Shani, S. N. Smith, B. J. Spring, L. M. Collins, K. Witkiewitz, A. Tewari, and S. A. Murphy, “Just-in-time adaptive interventions (JITAIs) in mobile health: Key components and design principles for ongoing health behavior support,” Ann. Behav. Med., vol. 52, no. 6, pp. 446–462, 2018, doi: 10.1007/s12160-016-9830-8.

[12] G. W. Fincham, C. Strauss, J. Montero-Marin, and K. Cavanagh, “Effect of breathwork on stress and mental health: A meta-analysis of randomised-controlled trials,” Sci. Rep., vol. 13, p. 432, 2023, doi: 10.1038/s41598-022-27247-y.

[13] R. Shao, I. S. C. Man, and T. M. C. Lee, “The effect of slow-paced breathing on cardiovascular and emotion functions: A meta-analysis and systematic review,” Mindfulness, vol. 15, no. 1, pp. 1–18, 2024, doi: 10.1007/s12671-023-02294-2.

[14] P. Miri, E. Jusuf, H. Margarit, R. Flory, A. Uusberg, K. Isbister, K. Marzullo, and J. J. Gross, “Evaluating a personalizable, inconspicuous vibrotactile (PIV) breathing pacer for in-the-moment affect regulation,” in Proc. CHI Conf. Human Factors Comput. Syst., 2020, doi: 10.1145/3313831.3376757.

[15] J. Costa, A. T. Adams, M. F. Jung, F. Guimbretière, and T. Choudhury, “EmotionCheck: A wearable device to regulate anxiety through false heart rate feedback,” GetMobile, vol. 21, no. 2, pp. 22–25, 2017, doi: 10.1145/3131214.3131222.

[16] M. de Witte, A. Spruit, S. van Hooren, X. Moonen, and G.-J. Stams, “Effects of music interventions on stress-related outcomes: A systematic review and two meta-analyses,” Health Psychol. Rev., vol. 14, no. 2, pp. 294–324, 2020, doi: 10.1080/17437199.2019.1627897.

[17] K. Starcke, J. Mayr, and R. von Georgi, “Emotion modulation through music after sadness induction—The iso principle in a controlled experimental study,” Int. J. Environ. Res. Public Health, vol. 18, no. 23, p. 12486, 2021, doi: 10.3390/ijerph182312486.

[18] Y. Koh, J. Kim, K. Lee, Y. Jung, H. Hong, and U. Lee, “Exploring data-driven approaches to stress management: A systematic review of stress tracking, intervention, and system evaluation methods,” in Proc. CHI Conf. Human Factors Comput. Syst., 2026, doi: 10.1145/3772318.3791194.

[19] A. Ta, N. Salgin, C. Armstrong, K. P. Reindel, and F. Sasangohar, “Ride, track, and recover: Pilot randomized trial of a wearable digital self-management intervention during a veteran endurance-cycling program,” arXiv:2606.13529, 2026.

[20] K. K. Jetelina, R. Molsberry, L. Malthaner, A. Beauchamp, M. B. Cannell, T. Hall, E. Fowler, and L. Anderson, “Acceptability of a real-time notification of stress and access to self-help therapies among law enforcement officers,” BMC Public Health, vol. 22, p. 32, 2022, doi: 10.1186/s12889-021-12423-y.

[21] J. M. van Oostrum, “Exploring the impact of wearable devices on perceived stress and health anxiety: A mixed-methods approach,” B.Sc. thesis, Univ. of Twente, Enschede, The Netherlands, 2024. [Online]. Available: https://essay.utwente.nl/100047/

[22] P. Miri, R. Flory, A. Uusberg, H. Culbertson, R. H. Harvey, A. Kelman, D. E. Peper, J. J. Gross, K. Isbister, and K. Marzullo, “PIV: Placement, pattern, and personalization of an inconspicuous vibrotactile breathing pacer,” ACM Trans. Comput.-Hum. Interact., vol. 27, no. 1, pp. 1–44, 2020, doi: 10.1145/3365107.

[23] X. Lowe-Brown, S. Glasser, G. Wadley, and P. Koval, “Personalised affect-regulation playlists: A pre-registered experimental test of the iso principle in the general population,” Musicae Scientiae, vol. 30, no. 3, pp. 428–446, 2026, doi: 10.1177/10298649261421187.